

Functional hard-sphere fluctuations on regular kinetic intervals

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Abstract

We prove functional Gaussian convergence of exactly centered hard-sphere fluctuations throughout a prescribed regular Boltzmann interval. The limit is a Radon law on the path space of tempered distributions, with the generalized Skorokhod topology induced by the strong dual, and is supported on strongly continuous paths. We use the explicitly stated analytic history package and finite-dimensional limit of the pinned upstream manuscript. New quantitative first-recollision estimates control connected cyclic histories by $C_h \mu^{1-h} \sqrt{\varepsilon(1 + \log(1/\varepsilon))}$, where $\mu = \varepsilon^{-2}$. A joint squared-flux argument preserves the source's factorial summability, including all permitted later weights. Actual pasted-law increment cumulants follow after microscopic remainder inversion. Passive collision records and a calendar-shift comparison give a microscopic modulus, closing a two-grid tightness argument. The true flow is reached by uniform exact-centered path comparison. No quantitative rate of convergence of the fluctuation law or its limiting covariance is asserted.

Dependency and manuscript status. The analytic history construction, its coarse estimates, and the existing finite-dimensional Gaussian limit are imported from the OpenAI preprint [1], at the exact snapshot specified below. The new Gaussian geometry and weighted summation are proved here. This is a research draft (version 1); external peer review and a comprehensive publication-priority review have not been completed. The strong-dual path topology is specified explicitly; convergence in a fixed negative-Sobolev path norm is not asserted. No publication-priority claim is made.

1 Model and results

Write $z = (x, v) \in \mathbb{R}^3 \times \mathbb{R}^3$. Define

$$\|g\|_{\text{Bol},\beta} = \sum_{a \in \mathbb{Z}^3} \sup_{|x-a| \leq 1, v \in \mathbb{R}^3} e^{\beta|v|^2} |g(x, v)|.$$

Assumption 1.1. *The initial datum f_0 is a smooth nonnegative probability density with $\|f_0\|_{\text{Bol},2\beta} + \|\nabla_x f_0\|_{\text{Bol},2\beta} < \infty$. On a fixed interval $[0, T]$ there is a nonnegative classical Boltzmann solution f with $f|_{t=0} = f_0$ satisfying $\sup_{t \leq T, x, v} e^{2\beta|v|^2} f_t(x, v) < \infty$.*

The scattering convention is

$$B(v - v_*, \omega) = [(v - v_*) \cdot \omega]_+, \quad v' = v - [(v - v_*) \cdot \omega]\omega, \quad v'_* = v_* + [(v - v_*) \cdot \omega]\omega,$$

with ordinary surface measure on S^2 , and $Q(f, f) = \int B(f'f'_* - ff_*) dv_* d\omega$. For diameter ε put $\mu = \varepsilon^{-2}$ and use the grand-canonical law

$$\mathbb{P}_\varepsilon(N, dZ_N) = Z_\varepsilon^{-1} \frac{\mu^N}{N!} \prod_{j=1}^N f_0(z_j) \prod_{i < j} \mathbf{1}_{\{|x_i - x_j| > \varepsilon\}} dZ_N.$$

The true flow has elastic hard-sphere collisions. We also use the single restarted pasted flow of [1, §3]: in each fixed slab, a proposed collision is retained only while its connected component remains below the prescribed size and cycle-excess caps. The parameter Γ and the number L of slabs are fixed before choosing any moment order. The size cutoffs are $A_L = \lfloor \log \varepsilon \rfloor$, $\Lambda_i = A_i^{30}$, and $A_{i-1} = \Lambda_i^{30}$, with harmless integer rounding. A proposed C contact is accepted only if the new component has at most Λ_i labels and excess at most Γ , with repeated edges counted. Otherwise the lines continue straight. Restarting changes only prehistory, not positions or velocities; overlaps are allowed for this auxiliary flow. Set $\tau = T/L$ and, for real $\phi \in C_c^\infty(\mathbb{R}^6)$,

$$\zeta_t^{\varepsilon, p}(\phi) = \mu^{-1/2} \left(\sum_{i=1}^N \phi(z_i^p(t)) - \mathbb{E} \sum_{i=1}^N \phi(z_i^p(t)) \right).$$

At every collision time we take the post-collision, right-continuous velocity version, for both true and pasted flows. Positions are continuous. The imported fixed-time laws are unchanged by this convention, as justified below. Every centering is the exact expectation of the indicated microscopic law.

Lemma 1.2 (Path convention). *Replacing incoming values at collision instants by outgoing values gives cadlag empirical test paths and changes neither a deterministic-time law nor its expectation.*

Proof. Conditional on each finite N , the initial law is absolutely continuous. The fixed-time collision sets are Lebesgue-null under the hard-sphere and capped flows; the latter preserve absolute continuity through their countable injective volume-preserving branches [2, Appendix A, proof of Proposition 4.5, items (1) and (5)]. Induct over the fixed slab restarts for the pasted law. At each fixed time both conventions consequently agree almost surely. A finite list of fixed times has the same joint law. The almost-everywhere flows have the usual one-sided collision limits; assigning the post-collision values gives cadlag paths. For bounded tests the sum is dominated by a constant times N , whose expectation is finite, so the mean is cadlag by dominated convergence. $\square$

Put $r_\varepsilon = \sqrt{\varepsilon(1 + \log(1/\varepsilon))}$ for $0 < \varepsilon \leq e^{-1}$.

Theorem 1.3 (Quantitative cyclic histories). *Under Assumption 1.1 and the imported package in Proposition 2.1, fix a factorial order h , a bounded finite record list, and a complexity cutoff K . The absolute sum of retained connected histories with trivial initial cumulants and at least one redundant contact is at most $C_{K,h,T} \mu^{1-h} r_\varepsilon$. The sum includes all sizes, all slab placements, and all permitted post-first-return weights. The repeated-earlier-edge sector has the stronger bound $C_{K,h,T} \mu^{1-h} \sqrt{\varepsilon}$. One sufficiently fine fixed slab decomposition works at every fixed K, h .*

Theorem 1.4 (Microscopic increment cumulants). *For the genuine pasted law and every fixed integer $k \geq 2$,*

$$|\kappa_k(\zeta_t^{\varepsilon, p}(\phi) - \zeta_s^{\varepsilon, p}(\phi))| \leq C_{k, \phi} \varepsilon^{k-2} (|t - s| + r_\varepsilon), \quad 0 \leq s \leq t \leq T.$$

The same dynamics is used at all orders. The sufficiently small upper bound on ε may depend on the fixed order. In particular $r_\varepsilon \leq C_\eta \varepsilon^\eta$ for every fixed $\eta < 1/2$.

Bodineau, Gallagher, Saint-Raymond and Simonella [3] already proved nonequilibrium Gaussian-process convergence on a short time interval. Their later work [4] treats arbitrarily long times at equilibrium. The source theorem [1] treats a regular kinetic lifespan away from equilibrium, but explicitly concerns finite-dimensional laws and does not assert path-space tightness. Gaussian-process convergence itself is therefore not the novelty at issue. The functional extension below concerns the source’s precise regular-lifespan hypotheses.

Strong distribution-valued conclusion. The principal functional result, Theorem 8.1, gives weak convergence of the centered empirical signed measures in $D([0, T], \mathcal{S}'_\beta(\mathbb{R}^6))$, for the generalized J_1 topology induced by the strong dual of the Schwartz space. Its limit is Radon and strongly continuous. The proof includes the extension to every Schwartz test and the path-law measurability and identification needed for the nuclear-space tightness theorem. No rate of convergence of these path laws or of their covariance to the Boltzmann limit is claimed.

2 The imported history package

We state the analytical and combinatorial inputs from [1], always at the exact commit specified in the bibliography. Their proofs, including the positive history estimates and kinetic construction based on [2], are not reproduced. Our results are conditional on this inherited package; we add no equilibrium, small-data, or shortened-lifespan assumption.

For a fixed bounded record attached to each particle, write ξ_i for its phase point together with that record. The ordered normalized factorial measure of the actual pasted law is

$$F_h^\varepsilon = \mu^{-h} \mathbb{E} \sum_{i_1, \dots, i_h \text{ distinct}} \delta_{(\xi_{i_1}, \dots, \xi_{i_h})}.$$

Its factorial cumulants are the finite partition polynomials

$$c_h^\varepsilon = \sum_{\pi \in \mathcal{P}([h])} (-1)^{|\pi|-1} (|\pi| - 1)! \bigotimes_{B \in \pi} F_{|B|}^\varepsilon,$$

with tensor factors assigned to their labeled slots. A hat denotes the source’s finite retained hierarchy and its corresponding cumulants. The source retains layered C/O histories with the cluster caps above, forest O-graphs between C-components, rooted layer components, forest initial links, and the continuation rules of its Definition `def:dhm-history`. Its complexity is $\rho = |H_0| + \sum_i (|H_i| + r_i)$: H_i is the set of designated or continuing roots in slab i , H_0 consists of lines incident to initial links, and r_i is the slab’s cycle excess. “Fixed complexity” means $\rho \leq K$. Thus it does not bound the total particle count. A global vertex is one physical particle line. A continued line retains its old phase coordinate; a new interface birth has an independent one. A contact between previously disjoint chronological components is a merger; all others are redundant. C scatters and O continues straight. At complexity K , let m count birth lines, n_i lines in slab i , a_i its mergers, q interface continuations and p physical initial links. For a connected history,

$$a = \sum_i a_i = m - 1 - p, \quad n = \sum_i n_i = m + q, \quad \sum_i (n_i - a_i) = q + 1 + p = O_K(1). \quad (2.1)$$

The redundant-contact count is also $O_K(1)$.

Proposition 2.1 (Imported package). *The following hold for the single pasted dynamics.*

- (i) *Independent birth inputs have one common Gaussian Bol envelope. Integrating a common spatial translation per component and taking spatial suprema of the other inputs gives $C^m e^{-bE}$, uniformly in every displacement, where E is total birth energy. Active energy never exceeds E .*
- (ii) *Merger coordinates consist of birth velocities, representative translations, an initial displacement per physical link, and a free (t, ω) per dynamic merger. Its Jacobian is $\varepsilon^2 B dt d\omega$. A redundant incoming intersection has indicator mass and a determined velocity update. A missing branch is extended by zero. Every step uses only preceding parameters.*
- (iii) *The normalization is $\mu^{m-h} \varepsilon^{2a} \varepsilon^{3p} = \mu^{1-h} \varepsilon^p$. Beyond ordered merger pair lists, choices cost $C_K(1+n)^{d_K} C^a$. Fresh-label denominators give $\prod_i n_i!$ in the denominator up to fixed polynomial factors. Merger pairs are not included in that prefactor.*
- (iv) *The actual marked factorial hierarchy and its retained version obey*

$$\|F_h^\varepsilon - \widehat{F}_h^\varepsilon\|_{\text{TV}} \leq C_h e^{-c_h \Lambda_L}, \quad \Lambda_L = (\log(1/\varepsilon))^{30}.$$

This is uniform at every fixed order and bounded record count, including partial final slabs. It is unweighted total variation. Retained factorial-cumulant inversion keeps exactly connected diagrams.

- (v) *For every fixed order and prescribed P , a fixed complexity cutoff makes the unscaled high-complexity retained tail $O(\varepsilon^P)$. Physical links gain ε^P . Nonphysical initial tails can be erased and assigned arbitrary powers by an auxiliary depth choice. These choices do not change the microscopic dynamics.*
- (vi) *Actual normalized factorial masses are at most one, and every fixed moment of N/μ is uniformly bounded.*
- (vii) *For the same dynamics, the actual unmarked cumulants obey $\|c_{h,t}^\varepsilon\|_{\text{TV}} \leq C_h \mu^{1-h}$ at every fixed order. The exactly centered pasted observations at any fixed finite list of compact smooth tests and times converge to the Gaussian law described in Section 7. Its target is a continuous linear map $C_0(\mathbb{R}^6) \rightarrow L^2$ with $\sup_{t \leq T} \|G_t(\phi)\|_{L^2} \leq C \|\phi\|_\infty$. On the common-initial-data coupling, the probability that the true and pasted trajectories differ anywhere in $[0, T]$ is at most $C\varepsilon^{21/20}$.*

Exponential bases and the choice of L are independent of subsequent fixed orders and cutoffs; front constants and polynomial degrees may depend on them.

The following exact source locators distinguish the inputs. Labels refer to the TeX sources at the pinned snapshot, not to claims proved here.

- Items (i)–(iii): [1, §3], `thm:dhm-package`; [1, §4], `prop:one-root`; and [1, §6], `eq:sharp-input-bounds`, `lem:sharp-merger-coordinates`, `lem:sharp-factorials`, and `lem:sharp-summability`.
- Items (iv)–(v): [1, §4], `lem:autonomous-partition`, `lem:one-slab-expansion`, `lem:marked-regrouping`, `lem:diagram-labels`, `lem:stopped-variation`, and `thm:marked-expansion`; physical-link scaling and initial-tail deletion are in [1, §6], `eq:sharp-initial-link` and `eq:sharp-scaling`.

- Item (vi): [1, §2], `prop:initial`, especially `eq:initial-factorial-upper` and `eq:initial-N-moments`.
- Item (vii): [1, §6], `thm:sharp-cumulants`; [1, §7], `prop:cumulant-clt`; [1, §2], `prop:gaussian-target` and `eq:target-field-bound`; and [1, §5], `prop:cutoff`. The exact-centering argument from [1, §8], `lem:center-transfer`, is reproduced below.

Required record lines continue to time zero. The finite identities depend on bounded output kernels and record count, not on a positive separation between observation times. Our Gaussian reference has covariance $\sigma^2 I$ per birth velocity, obtained after a fixed reservation of the exponent in (i). It is not a conditional law of the physical gas.

3 Gaussian geometry before the first redundancy

Lemma 3.1 (Transverse Gaussian bound). *Let (D, U) be jointly Gaussian in $\mathbb{R}^3 \times \mathbb{R}^3$, with U centered and isotropic. If $\lambda = \lambda_{\max} \text{Cov}(D | U) > 0$, then*

$$\mathbb{P}\{\text{dist}(D, \mathbb{R}U) \leq \varepsilon\} \leq C \min(1, \varepsilon/\sqrt{\lambda}).$$

Proof. Choose a top eigenvector e . Conditional on $U = u$, project onto $w = P_{u^\perp}e/\sin\theta$, where θ is the angle of u, e . The variance of $w \cdot D$ is at least $\lambda \sin^2\theta$. A one-dimensional Gaussian density gives $C \min(1, \varepsilon/(\sqrt{\lambda} \sin\theta))$, independently of its mean. The direction of U is uniform and $\mathbb{E}(\sin\theta)^{-1} = \pi/2$. The zero and parallel cases are null. $\square$

3.1 Repeated earlier pairs

Lemma 3.2 (An erased normal coordinate). *Fix a chronological merger forest, its normals and C/O flags. Suppose A and B have a prescribed contact at t_0 , and the prospective first redundant contact repeats this pair. Let $t_1 > t_0$ be the first later C contact touching either endpoint, and $\ell = t_1 - t_0$. At the beginning of the tested subsequent free segment, write D for their relative displacement and Y for both endpoint velocities. Then*

$$\text{Cov}(D | Y) \succeq \sigma^2 \ell^2 n_1 n_1^T, \quad \mathbb{P}_\gamma\{\text{return}\} \leq C_\sigma \min(1, \varepsilon/\ell).$$

If no such C contact exists, a second incoming crossing is impossible.

Proof. A C collision swaps the two normal components and retains the transverse ones. It is orthogonal in all birth Gaussian coordinates; the current velocity array is therefore a product Gaussian at fixed free normals. After A and B first belong to one component, every merger touching an endpoint imports a disjoint component. Its incident normal scalar is Gaussian and independent of the entire previous tagged history. That imported component cannot feed a scalar back through another merger: such a contact would already be redundant.

Suppose A has the first C update. With A_0, B_0 the velocities just after the earlier pair contact, set $\xi = A_0 \cdot n_1$. It is independent of $P_{n_1^\perp} A_0, B_0$ and all later imported scalars. At t_1 it is erased from A and never re-enters a tagged endpoint before the first redundancy. Consequently

$$D = \ell n_1 \xi + R, \quad \xi \text{ is independent of } (R, Y).$$

The initial contact displacement is a deterministic ε -normal offset. This proves the covariance inequality. Updating B changes only the sign. The final relative velocity is isotropic, and the proof of Lemma 3.1 gives the stated probability bound. Without a C update the relative line remains straight: a C contact separates, while an O contact has only the outgoing exit. Neither permits a second incoming crossing. $\square$

The conditioning is only on the two endpoint velocities, not on all partner velocities, realizability, or collision fluxes. The erased scalar was exported to a partner and can be revealed by those stronger conditionings.

3.2 Nonneighbor pairs

Lemma 3.3 (Rank one across a forest cut). *The 3 by 3 velocity-transfer block between distinct labels in a prescribed forest of C/O contacts has rank at most one.*

Proof. Cut an edge on their unique path. It is the only contact crossing that cut. A C contact transmits one normal scalar across it; O transmits none. Every later update is internal to one side, so dependence on the original velocity continues to factor through that scalar. This does not require the chronological order to be monotone along the spatial path. $\square$

Fix nonneighbors A,B. Orient tree edges $e = (p_e, q_e)$, put $b_e = e_{p_e} - e_{q_e}$, and write $b = e_A - e_B = \sum_e s_e b_e$ for their signed spatial path. The actual contact equations give, at a time t_* after the prefix,

$$D_\varepsilon = \varepsilon \sum_e s_e \omega_e + \sum_e s_e \int_{t_e}^{t_*} (v_{p_e}(r) - v_{q_e}(r)) dr. \quad (3.1)$$

This telescopes positions; it makes no constrained initial positions independent. At fixed normals all velocities are linear in the birth array V . Write $D_\varepsilon = \varepsilon \xi + L(t)V$, $U = QV$, $QQ^T = 2I$. Then

$$\text{Cov}(D_\varepsilon \mid U) = \sigma^2 H(t)H(t)^T, \quad H(t) = L(t) \left(I - \frac{1}{2} Q^T Q \right). \quad (3.2)$$

Velocity maps depend on the prescribed order and normals, not on the contact times inside their ordered domains.

Lemma 3.4 (A nonzero contact-time slope). *Let e_* be the earliest edge on the A–B spatial path. The derivative of the residual coefficient H with respect to its contact time has operator norm at least one.*

Proof. Before e_* the sum of the already-contacted path incidences is zero; just afterward it is $e_p - e_q$. Differentiating (3.1) gives the negative relative velocity immediately after e_* . Remove this edge from the final tree. The two sides have independent current Gaussian coordinates, and no later contact crosses the cut. If K_A, K_B transfer the current p, q velocities to the final A,B velocities, conditioning on both final velocities gives covariance

$$\sigma^2(2I - K_A^T K_A - K_B^T K_B).$$

At least one transfer is off-diagonal, because the path has at least two edges. It has rank at most one by Lemma 3.3 and norm at most one by orthogonality. On its kernel the displayed matrix has a direction of variance at least σ^2 . Conditioning only on the final difference increases covariance. Equation (3.2) proves the claim. $\square$

Proposition 3.5 (Time-averaged nonneighbor event). *Let a_i count all free merger times in the full schedule, including post-return mergers, and let i_* contain e_* . The Gaussian prefix return event G satisfies*

$$\int_{\prod_i \mathcal{S}_{a_i}(\tau)} \mathbb{P}_\gamma(G) dt \leq C \prod_i \frac{\tau^{a_i}}{a_i!} \min \left(1, \frac{a_{i_*} \varepsilon}{\sigma \tau} [1 + \log_+(\sigma \tau / \varepsilon)] \right).$$

This is uniform in all free normals and prescribed forest shapes.

Proof. Vary t_{e_*} between its nearest full-schedule neighbors in its slab, using slab endpoints if needed. Hold all other parameters fixed. The slope lemma supplies unit vectors w, z with $w^T H(t) z = \alpha t_{e_*} + \beta$, $|\alpha| \geq 1$. The largest conditional variance is at least $\sigma^2 |\alpha t_{e_*} + \beta|^2$. For an interval J of length s ,

$$\int_J \min(1, \rho/|\alpha t + \beta|) dt \leq 2\rho[1 + \log_+(s/(2\rho))], \quad \rho = \varepsilon/\sigma.$$

The even decreasing integrand has maximal interval integral when centered at its singularity. Apply Lemma 3.1, use $s \leq \tau$, and remove the selected time from its ordered simplex. The remaining volume is $\tau^{a_{i_*}-1}/(a_{i_*}-1)!$; every other slab factorial is unchanged. Also use the trivial probability bound.

For fixed birth and ending interfaces, allocate all birth Gaussian coordinates in advance, leave unborn coordinates unused, and freely extend terminated lines only for (3.1). The auxiliary extensions introduce no contact or random coordinate and cancel in the spatial telescoping sum. Varying the selected contact within its own slab retains the original interface assignments. $\square$

An untruncated inverse covariance moment at exponent one is not claimed: two O contacts on the path 1–2–3 give covariance proportional to the square of their time gap, whose inverse square root is not integrable. The truncated probability estimate legitimately retains one logarithm.

4 Fluxes and the complete retained size sum

Assume first that initial cumulants are trivial, so $p = 0$. Fix a bounded-complexity exceptional skeleton, including the first redundant-contact slot. Let s range over decorated merger pair schedules. After spatial-envelope domination write F_s for the full nonnegative merger-flux weight, with all needed redundant-branch indicators retained. It is zero on an unrealizable branch. Let G_s be the Gaussian transverse return demand determined solely by the formal forest before its first redundancy. Then $\text{supp } F_s \subset G_s$. No post-return Gaussian invariance is asserted.

Lemma 4.1 (Squared-flux sum). *At fixed merger times,*

$$\sum_s \int F_s^2 d\Omega \leq C^a E^a \prod_i n_i^{a_i}.$$

The number of ordered pair schedules is at most $C^a \prod_i n_i^{2a_i}$.

Proof. At a valid branch node in slab i ,

$$\sum_{p < q} |v_p - v_q|^2 = n_i \sum_p |v_p|^2 - \left| \sum_p v_p \right|^2 \leq n_i E.$$

The squared incoming flux is at most its squared relative speed. Integrate the free normal and sum C/O choices, then induct backward through the pair-choice tree. Every valid child preserves the energy bound, including all intervening determined redundant scatterings. Invalid children remain zero; adding other pair speeds only enlarges the estimate. Thus no fictitious velocity is needed on an invalid branch. The schedule count follows by permitting every unordered pair at every merger, ignoring forest restrictions only in this numerical upper bound. $\square$

Apply Cauchy–Schwarz jointly in schedules and the product parameter measure ν :

$$\sum_s \int F_s d\nu \leq \left(\sum_s \int F_s^2 d\nu \right)^{1/2} \left(\sum_s \int \mathbf{1}_{G_s} d\nu \right)^{1/2}. \quad (4.1)$$

For m birth Gaussians and $a = m - 1 \leq n$,

$$\mathbb{E}_\gamma E^a = (2\sigma^2)^a \frac{\Gamma(3m/2 + a)}{\Gamma(3m/2)} \leq (C_b n)^a.$$

In the nonneighbor case include *all* free merger times in ν . With $V_t = \prod_i \tau^{a_i}/a_i!$ and $B_\varepsilon = 1 + \log_+(\sigma\tau/\varepsilon)$, the two factors in (4.1), before taking square roots, are bounded by

$$C^{m+a} V_t n^a \prod_i n_i^{a_i}, \quad C^a V_t \prod_i n_i^{2a_i} \frac{C n \varepsilon B_\varepsilon}{\sigma\tau}.$$

Their product restores the full time volume and gives

$$C^{m+a} V_t n^{a/2} \prod_i n_i^{3a_i/2} \sqrt{\frac{n \varepsilon B_\varepsilon}{\sigma\tau}}. \quad (4.2)$$

Only the second factor uses the Gaussian forest geometry. In particular later merger weights and nonlinear determined returns stay in the first.

In the repeated-pair sector, refine the exceptional skeleton by the earlier pair-contact slot and the first later C endpoint-update slot. These at most $a^2 \leq n^2$ choices are absorbed in the fixed-complexity polynomial. Within each refined class t_0, t_1 , and the elementary gap selected below are common to all schedules. Classes with no such C update vanish by Lemma 3.2. We may now apply (4.1) at fixed times and use Lemma 3.2. The resulting factor is $\min(1, \sqrt{\varepsilon/(\sigma\ell)})$. If the earlier edge lies in slab i , choose either its next same-slab merger gap or the terminal gap to that slab's endpoint. In both cases the chosen gap Δ is at most ℓ . Under the normalized ordered slab simplex,

$$\mathbb{E} \Delta^{-1/2} = \tau^{-1/2} \frac{\Gamma(1/2) \Gamma(a_i + 1)}{\Gamma(a_i + 1/2)} \leq C \sqrt{(1 + a_i)/\tau}.$$

Thus (4.2) holds without B_ε in that sector.

Proof of Theorem 1.3. Insert the fresh-label denominator $\prod_i n_i!$ and the fixed complexity polynomial. The source's elementary Stirling calculation, with $\delta_i = n_i - a_i \geq 0$, gives

$$\frac{n_i^{3a_i/2}}{n_i! a_i!} \leq e^{2n_i} (1 + n_i)^{\delta_i} n_i^{-n_i/2}, \quad \prod_i n_i^{n_i} \geq (n/L)^n.$$

Because $\sum_i \delta_i = O_K(1)$, (4.2) is bounded by

$$C_{K,h,T} \mu^{1-h} r_\varepsilon (1 + n)^{d_K} (C_b \tau \sqrt{L})^n.$$

The activity factor is exactly $\mu^{m-h} \varepsilon^{2a} = \mu^{1-h}$. The number of slab-size vectors of total n is polynomial for fixed L . Choose L once so that $C_b T / \sqrt{L} < 1$. The size series converges. The repeated sector gives $\sqrt{\varepsilon}$ in place of r_ε . The first redundant pair is either an earlier edge or a nonneighbor, so these cases exhaust the $p = 0$ cyclic class. $\square$

Physical initial-link sectors separately have ε^p , $p \geq 1$, and the original summable majorant bounds them by $C_K \mu^{1-h} \varepsilon$. Nonphysical initial tails and high complexity are treated by their actual arbitrary-power estimates in Proposition 2.1; they are not made independent by the new geometry.

Remark 4.2. Uniform C^m flux moments for each fixed particle tree would be false. For collinear C contacts $(1, 2), (2, 3), \dots, (n-1, n)$, the same ancestral normal scalar is transported along the chain. Its flux product is $\prod_{j=2}^n (X_1 - X_j)_+$ with independent Gaussian X_j . The event $X_1/\sigma \in [2\sqrt{n-1}, 2\sqrt{n-1}+1]$ and $|X_j|/\sigma \leq 1$ gives expectation at least $c^n \sigma^{n-1} (n-1)^{(n-1)/2}$. The particle tree has degree two. Our all-schedule squared-flux proof preserves, and then cancels, this Gaussian growth through the two factorial denominators.

5 Actual microscopic increment cumulants

Fix $I = (s, t]$, and mark each designated particle by $y = \phi(z_t) - \phi(z_s)$ using two bounded observation pulses. Put $M = \|\phi\|_\infty$ and $D = \|v \cdot \nabla_x \phi\|_\infty$. The record satisfies $|y| \leq 2M$, and along every prescribed particle line

$$|y| \leq D|I| + 2MN_I, \quad (5.1)$$

where N_I may count all contacts of its diagram. Counting O contacts only increases this bound.

Lemma 5.1 (One-window tree localization). *For connected retained merger trees of factorial order h , fixed complexity and fixed total mark degree k , testing at least one increment factor gives an absolute bound $C_k \mu^{1-h}|I|$.*

Proof. For a ordered times in a slab of length τ ,

$$\int_{S_a(\tau)} \sum_{j=1}^a \mathbf{1}_{\{t_j \in I\}} dt = \frac{a|I \cap \text{slab}| \tau^{a-1}}{a!}.$$

This follows by symmetrizing over the cube. Before time integration the source's pair-flux and Gaussian majorants are uniform in the merger times. Inserting the count therefore costs only a polynomial size factor times $|I|/\tau$, preserving summability. Apply (5.1) to one marked factor and bound the others by $2M$. A contact-free term contributes only its streaming factor. Required record lines continue to time zero, so no part of this trajectory is hidden in an unmarked independent input. $\square$

Lemma 5.2 (Microscopic remainder after cumulant inversion). *For every fixed k and all $h \leq k$,*

$$\|c_h^\varepsilon - \tilde{c}_h^\varepsilon\|_{\text{TV}} \leq C_k e^{-c_k \Lambda_L},$$

uniformly in the two pulse times.

Proof. The actual normalized factorial masses are at most one. Thus the retained factorial variations are at most one plus their exponentially small errors. Invert the finite partition identity:

$$c_h = \sum_{\pi \in \mathcal{P}([h])} (-1)^{|\pi|-1} (|\pi| - 1)! \bigotimes_{B \in \pi} F_{|B|}.$$

Expand each tensor-product difference one factor at a time and use the factorial mass bounds. This proves the claim. Record updates are measurable pushforwards of norm at most one; their number and bounded ranges, not their time separation, control the source estimates. No velocity weight is attached to this error. $\square$

Proof of Theorem 1.4. Fix k and choose one complexity cutoff covering all $h \leq k$ with unscaled tail $O(\varepsilon^{2k+2})$. Choose the initial-tail depth afterward. Neither operation changes the dynamics. For integers $b_j \geq 1$ with $\sum_j b_j = k$, split retained connected diagrams into merger trees, cyclic $p = 0$ histories, physical-link histories and the initial-tail remainder. Lemma 5.1 and theorem 1.3 and the separate link gain imply

$$\left| \int c_h^\varepsilon \prod_{j=1}^h y_j^{b_j} \right| \leq C_k \mu^{1-h} (|I| + r_\varepsilon).$$

Here the actual microscopic error from Lemma 5.2 is smaller than every fixed power, and ε^{2k+2} is smaller than $\mu^{1-h} r_\varepsilon$ for $h \leq k$.

For $S_I = \sum_i y_i$, the finite marked partition identity is

$$\kappa_k(S_I) = \sum_{\pi \in \mathcal{P}([k])} \mu^{|\pi|} \int c_{|\pi|}^\varepsilon \prod_{B \in \pi} y_B^{|B|}.$$

It follows by expanding $e^{\lambda S_I} = \prod_i [1 + (e^{\lambda y_i} - 1)]$ and taking the logarithm; through degree k it is a finite formal identity. At fixed ε it is also justified by the exponential moments of N . For $k \geq 2$, exact centering changes no cumulant and each partition has normalization $\mu^{-k/2} \mu^{|\pi|} \mu^{1-|\pi|} = \mu^{1-k/2} = \varepsilon^{k-2}$. This proves the theorem, including uniformity over pulse times. The size-summation bases are independent of fixed order, so Γ, L remain the same throughout. $\square$

Corollary 5.3 (Even moments). *For each fixed integer $p \geq 1$, with $B_I = |I| + r_\varepsilon$,*

$$\mathbb{E}|\Delta_I \zeta^{\varepsilon, p}(\phi)|^{2p} \leq C_p (B_I^p + \varepsilon^{2p-2} B_I).$$

If $0 < a < 1/2$ and $|I| \leq \varepsilon^a$, the bound is $C_p \varepsilon^{ap}$.

Proof. A centered moment partition has no singleton blocks. A partition into $r \leq p$ blocks contributes $C_p \varepsilon^{2p-2r} B_I^r$. This geometric progression is bounded by a constant times its two endpoint terms. For the last assertion, $r_\varepsilon = o(\varepsilon^a)$ and $2p - 2 + a \geq ap$. $\square$

This argument proves a result for the actual pasted probability law, not only a signed retained expansion. It does not transfer its high moments to the true flow. The latter transfer will use uniform path comparison, which requires substantially less.

6 Collision counts and microscopic windows

We prove the needed source extensions explicitly. In this section a constant may depend on a requested fixed remainder accuracy.

Lemma 6.1 (A passive incidence record). *The exact component expansion and unweighted variation estimates in Proposition 2.1 also hold when a designated particle records its number of accepted C contacts in an interval I .*

Proof. In slab i an accepted component has at most $\Lambda_i - 1 + \Gamma$ edges. Thus each particle's total incidence record is at most $B_\varepsilon = \sum_i (\Lambda_i - 1 + \Gamma)$, a fixed power of $|\log \varepsilon|$. Normalize it by B_ε . In the source's autonomous partition identity, compatible autonomous components reproduce the full accepted root history, hence this statistic. The one-slab root-output kernel can therefore append it as one passive coordinate. Spectator cancellation and the fresh-label identity are unchanged. Pushforward has variation norm at most one; the crude operator bound and stopped forest domination do not require derivatives or a particular choice of output statistic. Required record labels still continue to zero, while unmarked one-root blocks are the same inputs g_i . Thus the source's finite regrouping and coarse estimates apply with one bounded coordinate per required label, not one coordinate per mesh point. At total degree k , restoring the raw record costs B_ε^k on errors. This is polylogarithmic, so exponentially small errors remain smaller than every power, and arbitrary-power tails can be chosen after k . On a retained diagram, raw mark powers cost only a fixed power of its contact count, hence a fixed polynomial in size at complexity K . $\square$

Lemma 6.2 (Calendar localization). *For fixed factorial order, complexity and total contact-mark degree, the sum of connected retained absolute marked integrals is at most $C \mu^{1-h} |I|$, provided at least one marked contact is required to lie in I . Nonphysical initial tails may instead be assigned arbitrary-power errors.*

Proof. First integrate out every common spatial translation and replace the birth factors by their uniform envelope $C^m e^{-bE}$. The remaining quotient coordinates are (V, Ω, t, y) ; there is no remaining integration in an unrestricted translation variable. Collapse identity interface gluings of continued lines. For the positive comparison family, freely extend each global line before its first and after its last prescribed contact. Redundant-contact segments end at neighboring prescribed contacts or at infinity, not at fixed physical lifetime interfaces. Keep the full chronological slot order and the requirement that each redundant incoming intersection exists; otherwise its branch weight is zero.

For $u \geq 0$ set

$$t_e \mapsto t_e + u, \quad y_{pq} \mapsto y_{pq} - u(v_p^{\text{birth}} - v_q^{\text{birth}}),$$

leaving birth velocities and free normals fixed. This quotient-coordinate map has determinant one. Reconstructing virtual paths inductively shows that the new path at time t is the old path at $t - u$, up to the already integrated representative translation. Merger normals, velocities and fluxes are unchanged. A redundant incoming intersection shifts by u with its normal and update unchanged. The first shifted contact is later than the original one, so its fixed birth anchor remains on its incoming free segment. Endings and nominal slab interfaces are only constraints in the original domain; virtual comparison paths may extend beyond them. All global coordinates remain in the energy accounting, and every update preserves total birth energy.

Put all absolute birth, ending, interface and nominal-slab restrictions, including those on determined redundant times, into a domain D . The Gaussian, fluxes and virtual chronological branch indicators form a calendar-invariant core W . If t_* is a selected contact, $t_*(\Phi_u \theta) = t_*(\theta) + u$. Changing variables, enlarging all shifted domains to D_ℓ^+ , and averaging over $0 \leq u \leq \ell$ gives

$$\int_D \mathbf{1}_{\{t_* \in I\}} W \, d\theta \leq \frac{|I|}{\ell} \int_{D_\ell^+} W^+ \, d\theta, \quad (6.1)$$

because $\int_0^\ell \mathbf{1}_{\{t_* \in I+u\}} \, du \leq |I|$. The changed interval $I + u$ is essential.

If $p = 0$, take $\ell = \tau$. Nominal merger-time intervals have length at most 2τ , and remain ordered within each slab. Their volumes are $(2\tau)^{a_i}/a_i!$. Retain the original global slot order until the final volume upper bound. The same pair-speed, Gaussian-energy and factorial estimates apply to these virtual histories: the subset of n_i nominal labels has $\sum_{p < q} |v_p - v_q| \leq n_i^{3/2} \sqrt{E}$. Thus their size majorant is $C_K \mu^{1-h} (1+n)^{d_K} (2C_b \tau \sqrt{L})^n$. Taking L sufficiently large once absorbs the enlargement.

If $p \geq 1$, take $\ell = \varepsilon < \tau$. An initial-link ball enlarges to radius $\varepsilon(R_{K,D} + \sqrt{2E})$. The p link volumes are at most $C_{K,D} \varepsilon^{3p} (1+E)^{3p/2}$. The normalization remains $\mu^{1-h} \varepsilon^p$. Averaging in (6.1) pays ε^{-1} ; the link gain leaves $\varepsilon^{p-1} \leq 1$. Absorb the fixed energy polynomial into half the birth Gaussian: $(1+E)^{3p/2} e^{-bE} \leq C_{b,p} e^{-bE/2}$. The fraction reserved is fixed independently of p , so the exponential size base stays independent of complexity and moment order. Nonphysical initial tails are erased with their inherited power cost and chosen afterward. Selecting a contact and restoring raw mark powers add only polynomial size factors. Summing the geometric size tail proves the lemma. $\square$

Proposition 6.3 (Actual incidence cumulants). *Let A_I be the total number of collision incidences, two per accepted C contact. For every fixed integers $k \geq 1$ and fixed accuracy $M > 0$,*

$$|\kappa_k(A_I)| \leq C_{k,M}(\mu|I| + \varepsilon^M)$$

uniformly in $I \subset [0, T]$.

Proof. Apply Lemma 6.1. In a connected retained term of total mark degree k , a nonzero mark product selects at least one C contact in I . Lemma 6.2 bounds its factorial-order- h contribution

by $C_{h,K,k}\mu^{1-h}|I|$. The ordinary-cumulant partition identity then multiplies by μ^h , giving $C\mu|I|$. Choose complexity, initial tails and stopped-error powers after k, M , allowing for the fixed factors μ^h and B_ε^k . The actual hierarchy-to-cumulant remainder is handled exactly as in Lemma 5.2. The leading constant may depend on M through K ; there is no assertion of a constant uniform over all accuracy choices. $\square$

Proposition 6.4 (Microscopic modulus). *For every fixed $J \geq 2$, $\lambda > 0$ and compact smooth ϕ ,*

$$\mathbb{P} \left\{ \sup_{|t-s| \leq \varepsilon^J} |\zeta_t^{\varepsilon,p}(\phi) - \zeta_s^{\varepsilon,p}(\phi)| > \lambda \right\} \leq C_{\phi,\lambda,J} \varepsilon^2$$

for sufficiently small ε .

Proof. Put $\delta = \varepsilon^J$ and cover the interval by $O(T/\delta)$ overlapping windows of length 2δ . The raw fourth-moment identity and Proposition 6.3, with the accuracy fixed sufficiently large after J , give

$$\mathbb{E}A_I^4 \leq C \sum_{r=1}^4 (\mu|I|)^r + O(\varepsilon^M).$$

For example accuracy $M+6$ in each cumulant absorbs the other at most three factors of order μ . Markov and the grid union bound yield

$$\mathbb{P}\{\max_I \varepsilon A_I > \lambda\} \leq C_\lambda(\varepsilon^2 + \delta + \varepsilon^{-2}\delta^2 + \varepsilon^{-4}\delta^3) + O(\varepsilon^{M+4-J}) = O(\varepsilon^2).$$

Pathwise, with $D_\phi = \|v \cdot \nabla_x \phi\|_\infty$,

$$\text{osc}_I \left(\varepsilon \sum_i \phi(z_i^p(t)) \right) \leq \varepsilon D_\phi N |I| + 2\|\phi\|_\infty \varepsilon A_I.$$

There are no state jumps when the pasted prehistory restarts. The single global N term needs no grid union: $\mathbb{E}N^2 \leq C\mu^2$ bounds its probability by $C\varepsilon^2\mu^2\delta^2 = C\varepsilon^{2J-2} \leq C\varepsilon^2$. The exact mean increment is deterministically at most $C_\phi\varepsilon^{-1}\delta + O(\varepsilon^M)$ by the first incidence cumulant. It is smaller than a fixed threshold for small ε . This proves the statement, with constants allowed to depend on J . $\square$

7 Finite families of fluctuation processes

For a test ϕ write $\Delta\phi = \phi(x, v') + \phi(x, v'_*) - \phi(x, v) - \phi(x, v_*)$. The inherited Gaussian finite-dimensional law has weak drift and noise

$$A_t\phi = v \cdot \nabla_x \phi + \int B(v - v_*, \omega) f_t(x, v_*) \Delta\phi \, dv_* \, d\omega,$$

$$C_t(\phi, \psi) = \frac{1}{2} \int B(v - v_*, \omega) f_t(x, v) f_t(x, v_*) \Delta\phi \Delta\psi \, dx \, dv \, dv_* \, d\omega.$$

Its initial covariance is $\int f_0 \phi \psi$. The initial field is independent of the Gaussian noise whose covariance is $\int_0^{s \wedge t} C_r(\phi, \psi) \, dr$. We use the source's backward mild construction and finite-dimensional identification [1, §§2,7–8], not an assumed path-space theorem.

Theorem 7.1 (Regular-lifespan finite-family convergence). *Under Assumption 1.1 and the imported package, for every fixed $\phi_1, \dots, \phi_d \in C_c^\infty(\mathbb{R}^6)$ the exactly centered true hard-sphere processes converge in $D([0, T], \mathbb{R}^d)$ with its Skorokhod J_1 topology to a continuous Gaussian version of the inherited finite-dimensional law. The same holds for the one fixed pasted dynamics.*

Proof. Choose Γ as required by the source's component expansion and global cutoff estimate. Choose L once large enough for its original majorants, the squared-flux bounds, and the doubled-slab/fixed-Gaussian reservation in Lemma 6.2. All additional bases are independent of moment order and auxiliary accuracy. Thus all estimates below concern one microscopic process. Accuracy cutoffs are chosen only afterward.

It suffices first to work with one fixed test, writing $X_\varepsilon(t)$ for its pasted fluctuation. The finite-particle paths are cadlag. Their expectation is cadlag by dominated convergence, since the observation sum is bounded by $\|\phi\|_\infty N$ and $\mathbb{E}N < \infty$.

Use a uniform mesoscopic grid of spacing $h_\varepsilon \asymp \varepsilon^{1/4}$ and let Y_ε linearly interpolate the values of X_ε there. For nodes separated by $d \geq h_\varepsilon$, Corollary 5.3 gives

$$\mathbb{E}|X_\varepsilon(t) - X_\varepsilon(s)|^4 \leq C[(d + r_\varepsilon)^2 + \varepsilon^2(d + r_\varepsilon)] \leq C'd^2.$$

The same bound $C|t - s|^2$ holds for the interpolation at arbitrary times: inside a cell its fourth moment has the fractional factor $(|t - s|/h_\varepsilon)^4$, and across cells split into the two end pieces and the node-to-node middle interval. Tight initial laws are inherited. The Kolmogorov criterion therefore makes Y_ε tight in $C([0, T])$.

Subdivide each mesoscopic cell into equal intervals of length at most ε^2 and comparable to it. There are $O(\varepsilon^{-2})$ fine-grid points. For each point compare X_ε with its mesoscopic left endpoint, also including right endpoints in the comparison. The fixed eighteenth-moment case of Corollary 5.3 gives

$$\mathbb{P} \left\{ \max_{\text{fine points } t} |X_\varepsilon(t) - X_\varepsilon(\ell(t))| > \lambda \right\} \leq C_\lambda \varepsilon^{9/4-2} \rightarrow 0.$$

In particular all neighboring mesoscopic increments vanish uniformly. Proposition 6.4 with $J = 2$ controls every point between fine nodes. It follows that $\sup_{t \leq T} |X_\varepsilon(t) - Y_\varepsilon(t)| \rightarrow 0$ in probability. Fixed-time comparisons and the inherited finite-dimensional Gaussian convergence identify every subsequential interpolation limit. This also constructs a continuous version of that law. A finite product argument gives the joint assertion for the d tests.

Finally couple true and pasted flows by their common initial data. The source cutoff event E_ε concerns disagreement anywhere in $[0, T]$ and has probability $O(\varepsilon^{21/20})$ [1, §5]. Both flows conserve N . Uniformly in time,

$$\sqrt{\mu} |\mathbb{E}\pi_t^{\varepsilon, \text{true}}(\phi) - \mathbb{E}\pi_t^{\varepsilon, p}(\phi)| \leq 2\|\phi\|_\infty \varepsilon^{-1} \mathbb{E}[(N/\mu)\mathbf{1}_{E_\varepsilon}] \leq C_\phi \varepsilon^{1/40},$$

using the uniform forty-second moment and $(21/20)(41/42) - 1 = 1/40$. Outside E_ε the uncentered paths agree at every time. Hence the exactly centered paths differ uniformly in probability by something tending to zero. This transfers the finite-family weak limit. $\square$

Remark 7.2. Only finite families of tests are asserted in Theorem 7.1. The separate strong-dual functional theorem, including its topology and tightness argument, is proved in Theorem 8.1. Likewise uniform path comparison does not imply true-flow high-moment convergence: a fluctuation of size ε^{-1} on an event of probability $\varepsilon^{21/20}$ has diverging variance despite vanishing probability. None of our conclusions uses that invalid transfer.

8 The strong tempered-distribution path limit

Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^6)$ be the real Schwartz space and let $\mathcal{S}'_\beta$ be its dual with the strong topology. Thus the state-space seminorms are

$$p_B(u) = \sup_{\phi \in B} |u(\phi)|, \quad B \subset \mathcal{S} \text{ bounded.}$$

The family is directed by finite unions and positive rescaling. We use the generalized Skorokhod J_1 topology on the space of strongly cadlag paths, generated by

$$d_B(x, y) = \inf_{\lambda \in \Lambda} \max \left\{ \sup_{t \leq T} |\lambda(t) - t|, \sup_{t \leq T} p_B(x(\lambda(t)) - y(t)) \right\}, \quad (8.1)$$

where Λ is the set of increasing homeomorphisms of $[0, T]$ onto itself. This is the strong-state version of [6, §1, equation (1.3)], after time rescaling. It is not defined merely as a product of scalar path topologies. For each $\phi \in \mathcal{S}$, the map

$$\Pi_\phi : D([0, T], \mathcal{S}'_\beta) \longrightarrow D([0, T], \mathbb{R}), \quad (\Pi_\phi x)(t) = x(t)(\phi),$$

is continuous by (8.1), using the bounded set $B = \{\phi\}$. The same observation applies to finite vectors of tests.

For either microscopic dynamics define the signed measure

$$Z_t^\varepsilon = \mu^{-1/2} (U_t^\varepsilon - \mathbb{E} U_t^\varepsilon), \quad U_t^\varepsilon = \sum_{i=1}^N \delta_{z_i(t)}. \quad (8.2)$$

The paths use the post-collision convention of Lemma 1.2. The expectation in (8.2) can equivalently be taken as a Bochner expectation in the auxiliary Hilbert space used below.

Theorem 8.1 (Strong tempered-distribution convergence). *Under Assumption 1.1 and the imported package of Proposition 2.1, the laws of (8.2), for both the true and the one fixed pasted dynamics, converge weakly in $D([0, T], \mathcal{S}'_\beta)$ with topology (8.1). The common limit is a Radon probability law supported on $C([0, T], \mathcal{S}'_\beta)$. Its finite-dimensional evaluations on Schwartz tests are the centered Gaussian field of the source, restricted from $E_0^0 = C_0(\mathbb{R}^6)$ to $\mathcal{S}$. Its covariance is*

$$\begin{aligned} \mathbb{E}[G_t(\phi)G_s(\psi)] &= \int f_0(z)U(0, t)\phi(z)U(0, s)\psi(z) \, dz \\ &\quad + \int_0^{s \wedge t} C_r(U(r, t)\phi, U(r, s)\psi) \, dr, \end{aligned} \quad (8.3)$$

where $U(r, t)$ is the source's backward propagator and C_r is the collision covariance from Section 7.

Here weak convergence tests all bounded continuous functions on the specified path space. Neither tightness in a fixed negative-Sobolev norm nor convergence for the uniform topology on all discontinuous paths is asserted. On the continuous-path subspace, the topology induced by (8.1) is uniform convergence in every strong-dual seminorm. We prove all test-class, path-realization, and law-identification steps before concluding the theorem.

8.1 The all-Schwartz scalar estimates

Lemma 8.2. *For every $\phi \in \mathcal{S}$ the pasted scalar fields $Z^{\varepsilon, p}(\phi)$ are C -tight in $D([0, T], \mathbb{R})$: they are tight and all subsequential limits are supported on continuous paths. At every fixed order $k \geq 2$, their increment cumulants satisfy*

$$|\kappa_k(Z_t^{\varepsilon, p}(\phi) - Z_s^{\varepsilon, p}(\phi))| \leq C_k q(\phi)^k \varepsilon^{k-2} (|t - s| + r_\varepsilon), \quad q(\phi) = \|\phi\|_\infty + \|v \cdot \nabla_x \phi\|_\infty. \quad (8.4)$$

The choice of pasted dynamics does not depend on ϕ or k .

Proof. Both terms defining q are finite, and q is a continuous seminorm on $\mathcal{S}$. We check explicitly why compact support was not used in the estimates of Theorem 1.4 and proposition 6.4.

The source's marked convention permits bounded continuous observation functions [1, §6, Conventions and the assertion]. A pulse $y \mapsto y + \phi(z)$ is a measurable output pushforward. In the component kernels, stopped variation estimates, and bounded record factors of Proposition 2.1, only the record range and the fixed number of record-bearing labels enter. A pushforward does not increase total variation. Normalizing a nonzero test by its supremum norm keeps a two-pulse record in $[-2, 2]$; homogeneity restores the original scale. No integration over the support of ϕ occurs. The spatial integrals instead use a birth density in each component translation and the fixed envelopes of the other birth densities.

Put $M = \|\phi\|_\infty$ and $D = \|v \cdot \nabla_x \phi\|_\infty$. On every designated particle line of an autonomous or retained history, the fundamental theorem of calculus between contacts and the bounded jump estimate give

$$|\phi(z(t)) - \phi(z(s))| \leq D(t - s) + 2MN_{(s,t]},$$

where the contact count may be enlarged to include the entire diagram. The localized tree proof uses this inequality for one increment factor and bounds the other factors by $2M$. Its contact-count insertion costs only a fixed polynomial in diagram size. The cyclic, initial-link, high-complexity, and actual microscopic-error terms drop bounded marks. Thus normalization by $q(\phi)$ proves (8.4) by the same proof as Theorem 1.4; the case $q(\phi) = 0$ is immediate. The diagram bases, and hence Γ, L , are unchanged.

The count A_I in Proposition 6.4 is independent of ϕ . Its conversion to an observable uses only

$$\text{osc}_I \left(\varepsilon \sum_i \phi(z_i(t)) \right) \leq \varepsilon DN|I| + 2M\varepsilon A_I.$$

For the mean use the same inequality with expectations. The bound $\mathbb{E}N^2 \leq C\mu^2$ and the count estimates therefore give the same microscopic modulus for every Schwartz test. The initial CLT in [1, §2, Initial ensemble] includes bounded tests, so the initial scalar laws are tight. The moment-cumulant and two-grid interpolation proof of Theorem 7.1 now applies unchanged, including its fixed high-moment choice: the continuous interpolations are tight and their uniform distance from $Z^{\varepsilon,p}(\phi)$ vanishes in probability. This proves C -tightness for every element of $\mathcal{S}$, rather than only for a dense family of compact tests. $\square$

8.2 Identification beyond compact tests

Lemma 8.3. *Every finite collection of pasted Schwartz observations converges to the corresponding source Gaussian observations.*

Proof. At a fixed time t , take the actual unmarked normalized factorial cumulant measures $c_{1,t}^\varepsilon, c_{2,t}^\varepsilon$. The source's sharp connected bound gives

$$\|c_{1,t}^\varepsilon\|_{\text{TV}} + \mu \|c_{2,t}^\varepsilon\|_{\text{TV}} \leq C_t \quad [1, \S 6, \text{Sharp connected bounds}].$$

These are bounds on the unmarked measures, with constants independent of a subsequently chosen test. For any bounded measurable h the ordinary second-cumulant identity therefore yields

$$\mathbb{E}|Z_t^{\varepsilon,p}(h)|^2 = c_{1,t}^\varepsilon(h^2) + \mu c_{2,t}^\varepsilon(h \otimes h) \leq C_t \|h\|_\infty^2. \quad (8.5)$$

Only finitely many fixed times are at issue, so a time-uniform constant is unnecessary.

Choose $\chi \in C_c^\infty(\mathbb{R}^6)$ equal to one near zero and put $\phi_R(z) = \chi(z/R)\phi(z)$. For $\phi \in \mathcal{S}$, one has $\phi_R \in C_c^\infty$, $\|\phi_R - \phi\|_\infty \rightarrow 0$, and $\phi_R \rightarrow \phi$ in all Schwartz seminorms. For every finite observation vector, (8.5) and a finite sum show that the cutoff vector approximates the original one in probability uniformly in small ε as $R \rightarrow \infty$. Its fixed- R limit is identified by Theorem 7.1.

The source's Gaussian target is a continuous linear map $E_a^0 \rightarrow L^2$ uniformly in t , and at $a = 0$ this gives

$$\sup_{t \leq T} \|G_t(h)\|_{L^2} \leq C\|h\|_\infty, \quad h \in C_0(\mathbb{R}^6) \quad [1, \S 2, \text{Gaussian target}]. \quad (8.6)$$

Thus the Gaussian cutoff observations converge in L^2 . The elementary approximation theorem for weak convergence proves the assertion and identifies the covariance with (8.3). The argument used an actual unweighted microscopic covariance bound, not a weighted remainder estimate or a variance bound for the true law. $\square$

8.3 Strong paths, exact means, and the true-flow comparison

Lemma 8.4. *For every ε , each field in (8.2) defines a Radon Borel probability law on $D([0, T], \mathcal{S}'_\beta)$. For any fixed $s > 3$, the true and pasted fields can be coupled so that*

$$\sup_{t \leq T} \|Z_t^{\varepsilon, \text{true}} - Z_t^{\varepsilon, p}\|_{H^{-s}(\mathbb{R}^6)} \longrightarrow 0 \quad \text{in probability.} \quad (8.7)$$

Proof. Use the ordinary inhomogeneous Sobolev space $H^{-s}(\mathbb{R}^6)$. For $s > 3$, Fourier transformation gives, up to the fixed normalization,

$$\|\delta_z\|_{H^{-s}}^2 = c \int_{\mathbb{R}^6} (1 + |\xi|^2)^{-s} d\xi = C_s^2 < \infty.$$

Dominated convergence applied to $e^{-iz \cdot \xi}$ proves that $z \mapsto \delta_z$ is norm-continuous into this separable Hilbert space. The finite-particle flow has finitely many collisions outside its exceptional null set; see [2, Proposition 1.2 and Proposition 4.5]. For the pasted flow, the accepted-edge caps also give this directly in each of the finitely many slabs. Lemma 1.2 fixes the right-continuous version, and no phase jump is created by a restart or a rejected contact. Consequently U^ε is H^{-s} -cadlag and

$$\sup_{t \leq T} \|U_t^\varepsilon\|_{H^{-s}} \leq C_s N.$$

Since $\mathbb{E}N < \infty$, its fixed-time Bochner means exist. Dominated convergence shows that the mean is right-continuous and has left limit $\mathbb{E}U_{t-}^\varepsilon$ at each $t > 0$. Thus the exactly centered field is also H^{-s} -cadlag. This argument needs no mean-continuity assertion; in fact the fixed-time collision-null property in Lemma 1.2 and the same domination show that the mean is continuous.

The source flow is measurable at each deterministic time. Fixed-time measurability and cadlag paths make Z^ε a Borel random variable in the Polish space $D([0, T], H^{-s})$, whose Borel sigma-field is generated by its coordinates. We are not invoking continuity of subtraction in J_1 at common jumps. For a bounded Schwartz set B ,

$$p_B(u) \leq \left(\sup_{\phi \in B} \|\phi\|_{H^s} \right) \|u\|_{H^{-s}}. \quad (8.8)$$

The bracket is finite, since $\mathcal{S} \rightarrow H^s$ is continuous. Hence $H^{-s} \rightarrow \mathcal{S}'_\beta$ and its induced path inclusion are continuous, the latter directly by (8.1). The pushforward of a Radon law under this continuous

inclusion is Radon: for any Borel set, approximate its inverse image from inside by compacts and take their compact images. This proves the first claim.

For the second claim let E_ε be the source's full-time discrepancy event, used in Theorem 7.1. Outside it the uncentered paths agree, while always

$$\sup_t \|U_t^{\varepsilon, \text{true}} - U_t^{\varepsilon, p}\|_{H^{-s}} \leq 2C_s N \mathbf{1}_{E_\varepsilon}.$$

Subtracting the two Bochner means gives

$$\sup_t \|Z_t^{\varepsilon, \text{true}} - Z_t^{\varepsilon, p}\|_{H^{-s}} \leq 2C_s \mu^{-1/2} (N \mathbf{1}_{E_\varepsilon} + \mathbb{E}[N \mathbf{1}_{E_\varepsilon}]).$$

The deterministic term tends to zero by the exact-centering Hölder estimate in Theorem 7.1; the random term vanishes outside an event of probability $O(\varepsilon^{21/20})$. This proves (8.7). It transfers scalar C -tightness and the Schwartz finite-dimensional limits to the true flow by (8.8). It transfers no true-flow high moments and asserts no Hilbert-norm tightness of the family. $\square$

8.4 Nuclear tightness, Borel determination, and subsequences

We now apply Mitoma's theorem to the full Schwartz test space: on the strong dual of a nuclear Fréchet space, tightness of every scalar projection is equivalent to tightness of the path laws [5]. The precise strong-topology statement also appears in [6, §5.II, Theorem 5.5]. Its hypotheses hold here because $\mathcal{S}$ is nuclear Fréchet, Lemma 8.4 gives Borel path laws, and Lemma 8.2 covers every $\phi \in \mathcal{S}$. Therefore the pasted laws are tight along every sequence $\varepsilon_n \rightarrow 0$. The last paragraph of Lemma 8.4 permits the same application to the true laws. We do not need a nonmetrizable Slutsky theorem. The following details justify both subsequences and uniqueness.

Choose a countable $\mathcal{S}$ -dense subset $\{\phi_j : j \geq 1\} \subset C_c^\infty(\mathbb{R}^6)$, replacing it by its rational linear span if desired. Such a set exists by separability and the Schwartz cutoff approximation above. These tests separate all distributions.

We first record why countably many pairings determine the strong state-space Borel sigma-field. Take an increasing Hilbertian scale defining $\mathcal{S}$, with compact connecting maps at suitably spaced indices; Hermite-function norms provide such a scale. The dual is the union of the negative Hilbert spaces. Closed integer-radius balls in these spaces, viewed in sufficiently weaker spaces, are compact, and their images are compact in $\mathcal{S}'_\beta$. They form a countable cover $\{K_n\}$ of $\mathcal{S}'$. The coordinate map $u \mapsto (u(\phi_j))_j$ is a continuous injection into $\mathbb{R}^\mathbb{N}$ and is a homeomorphism on each K_n . A strong Borel set intersected with K_n therefore has a Borel coordinate image. The compact image of K_n is Borel in the product, and the global injection recovers that intersection by inverse image. Taking the countable union over n proves the state-space assertion.

The same Hilbert-scale argument explains the path-space hypothesis in [6, Proposition 5.3]. A strongly cadlag path is bounded on $[0, T]$ after every scalar pairing. Banach–Steinhaus gives one dual-Hilbert norm and one radius bounding its entire range, and a compact inclusion places the range in a compact ball of the preceding kind. Proposition 5.3(ii) gives equality of the strong path Borel and simple cylindrical sigma-fields. Combining it with the state-space fact and right continuity gives

$$\mathcal{B}(D([0, T], \mathcal{S}'_\beta)) = \sigma\{x \mapsto x(q)(\phi_j) : j \geq 1, q \in Q_T\}, \quad Q_T = ([0, T] \cap \mathbb{Q}) \cup \{T\}. \quad (8.9)$$

Indeed every nonterminal time is approached from above by rational times. This is a Borel-generation statement, not just separation of points.

For completeness, tightness supplies actual weakly convergent subsequences even though the strong path space is not Polish. Consider the continuous injection

$$J : D([0, T], \mathcal{S}'_\beta) \longrightarrow F = \prod_{j \geq 1} D([0, T], \mathbb{R}), \quad Jx = (\Pi_{\phi_j} x)_j.$$

The product F is Polish, and J is a homeomorphism on every compact subset of its domain. Given a tight sequence P_n , choose increasing compact sets L_m with $P_n(L_m) \geq 1 - 2^{-m}$ for every n . Ordinary Polish Prokhorov yields a subsequence for which $J_\# P_n \Rightarrow \nu$ on F . Since $J(L_m)$ is compact, portmanteau gives $\nu(J(L_m)) \geq 1 - 2^{-m}$. Thus ν is supported on the Borel set $A = \bigcup_m J(L_m)$. The inverse of J on A is Borel, being continuous on each compact piece. Its pushforward defines a probability P on the strong path space. This law is Radon: restrict first to one compact piece and use regularity of ν there to approximate every Borel set from within by compact sets, then let the piece increase.

If f is bounded continuous on the strong path space, transport its restriction to L_m to $J(L_m)$ and extend it by Tietze to a bounded continuous f_m on F , without enlarging its supremum norm. For both P_n and P , the difference between the desired integral and that of the extension is at most $2\|f\|_\infty 2^{-m}$. First use $J_\# P_n \Rightarrow \nu$, then let $m \rightarrow \infty$. This proves $P_n \Rightarrow P$ in the original strong topology and establishes a Radon subsequential limit without metrizing that topology.

8.5 Strong continuity and the unique limit law

Proof of Theorem 8.1. Let P be any subsequential limit obtained above, for either family. The continuous map Π_{ϕ_j} sends it to a scalar subsequential limit. Scalar C -tightness implies that $\Pi_{\phi_j} x$ is continuous for P -almost every x . Intersect these countably many probability-one events. Every remaining path is already strongly cadlag, and at every $t > 0$

$$(x(t) - x(t-))(\phi_j) = 0 \quad \text{for every } j.$$

Density of the tests implies $x(t) = x(t-)$ as distributions, for every $t > 0$ simultaneously. Equality to the strong left limit, together with strong right continuity, proves strong continuity, including at T . There is no uncountable union of null sets in this argument. The continuous-path set is Borel: it equals $\bigcap_j \Pi_{\phi_j}^{-1} C([0, T], \mathbb{R})$ on the already cadlag space. It is also closed, and its topology from (8.1) is uniform convergence in each p_B [6, Proposition 1.6(i)]. The latter statement also follows directly from the uniform p_B -continuity of a continuous limiting path and the time changes in (8.1).

Finite-vector scalar path projections are continuous into ordinary finite-dimensional J_1 spaces. Their limiting paths are continuous, so the usual finite-dimensional evaluation theorem identifies all their time marginals with Lemma 8.3; for the true law use (8.7) as already explained. In particular the finite-dimensional distributions of the coordinates in (8.9) have one prescribed Gaussian law. They determine at most one probability measure on that Borel sigma-field. Hence every subsequential limit of either family is the same P . Tightness and this uniqueness, applied to every sequence $\varepsilon_n \rightarrow 0$, give full convergence. Equation (8.3) follows from Lemma 8.3. This constructs the claimed Radon, strongly continuous realization of the source Gaussian target; it was not assumed as an input. $\square$

Remark 8.5 (Scope of the distribution-valued statement). Mitoma was applied only after scalar tightness had been proved for every Schwartz test. Compact tests alone were used for the original finite-dimensional identification and for a countable determining family. The actual microscopic errors remained unweighted and were tested only with bounded records. The source's Gaussian field is used with its existing test-scale interpretation: we do not claim that the collision drift maps $\mathcal{S}$

into $\mathcal{S}$, or enlarge the source's class of solutions in its uniqueness assertion. The Hilbert space H^{-s} in Lemma 8.4 is a device for individual path laws and coupling, not an asserted uniform tightness topology.

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